

Structural Thresholds of Stepping-Stone Graphs in Euclidean Space

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ABSTRACT

We study the structural properties of stepping-stone proximity graphs in $\mathbb{R}^d$, for every $d \geq 2$. The defining diversion neighborhoods form a strictly increasing family between the segment, the Gabriel ball, and the relative-neighborhood lune, so the associated graphs form an edge-decreasing filtration. We derive dimension-independent inclusion chains with classical proximity graphs and obtain two-sided comparisons with β -skeletons. These comparisons recover the known planar Delaunay construction and transfer connectivity, tiling-recovery, and spanning-ratio consequences.

The parameter $\alpha = 2$ is shown to be the sharp universal threshold, in every ambient dimension, both for intersection-free straight-line realizations and for containment in the Delaunay graph. Abstract planarity behaves differently: $\alpha = 2$ is the sharp threshold in the plane, whereas in every dimension $d \geq 3$ there are finite point sets whose stepping-stone graph is nonplanar for every parameter. Finally, we extend the planar stepping-stone spectrum to arbitrary dimension, give an exact site-entry equation with the strict edge convention, bound the number of distinct graphs in the filtration, and prove explicit finite-set stabilization at the open relative-neighborhood graph. The construction is available in Python through the open-source PROXIMITYGRAPHS package.

1. Introduction

An empty-region graph joins two sites whenever a region associated with the pair contains no other site. The Gabriel graph, relative-neighborhood graph, and β -skeletons are standard examples (Gabriel and Sokal, 1969; Toussaint, 1980; Jaromczyk and Toussaint, 1992; Kirkpatrick and Radke, 1985). Their structural properties are controlled by the geometry and boundary convention of the corresponding empty region. In the plane, Cardinal, Collette and Langerman (2009) formalized this dependence through convex and symmetric anchored regions and characterized tight regions for several universal graph properties.

The stepping-stone graph was introduced by Kannangara, Tanin, Harwood and Karunasekera (2018) and developed further by Kannangara, Tanin, Harwood and Karunasekera (2019) for extracting movement corridors from sparse trajectories. For distinct sites p, q , its diversion neighborhood at parameter $\alpha \geq 1$ consists of the points z satisfying

$$\|z - p\|^\alpha + \|z - q\|^\alpha \leq \|p - q\|^\alpha. \quad (1)$$

The two sites are adjacent when this region contains no other point of the input set. The original planar theory established graph monotonicity, the Gabriel and relative-neighborhood endpoints, planarity for $\alpha \geq 2$, a Delaunay-based construction, and the planar d -spectrum. The defining Euclidean inequality, however, is meaningful in every ambient dimension.

This paper investigates the resulting structural theory in $\mathbb{R}^d$. The central distinction is between properties of a particular straight-line realization and properties of the abstract graph. Two segments that intersect properly always lie in a common affine plane, so a planar obstruction can control geometric intersections in every dimension. Abstract planarity does not behave this way: higher-dimensional point sets may realize nonplanar graphs without any proper edge intersections.

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Our first contribution is a dimension-independent region-inclusion analysis. It yields the edge-decreasing stepping-stone filtration, its classical proximity-graph chains, and explicit comparisons with both lune-based and circle-based β -skeletons. These comparisons transfer connectivity, Delaunay, tiling-recovery, and spanning-ratio results while keeping known planar statements and their higher-dimensional extensions distinct.

Our second contribution is a sharp threshold theory. We prove that $\alpha = 2$ is the exact universal threshold, in every dimension, for intersection-free straight-line realizations and Delaunay containment. In the plane it is also the exact threshold for abstract planarity. In dimensions at least three, by contrast, finite integer-coordinate point sets produce nonplanar stepping-stone graphs for every parameter.

Finally, we extend the stepping-stone spectrum of Kannangara et al. (2018) to arbitrary dimension. An exact entry equation determines when a site first enters a pair's diversion neighborhood. For every finite point set, the resulting edge filtration changes only at finitely many parameter values and stabilizes explicitly at the open relative-neighborhood graph. The reusable construction is available in Python through PROXIMITYGRAPHS, complementing the earlier Java implementation reported by Kannangara et al.

The remainder of the paper is organized as follows. Section 2 introduces the regions, boundary conventions, and similarity normalization. Section 3 records the endpoint and limiting regions. Section 4 develops the inclusion theory, β -skeleton comparisons, sharp thresholds, dimension-dependent planarity, and finite-set spectrum. Section 5 describes the available Java and Python implementations.

2. Definitions and geometric preliminaries

Throughout, $d \geq 2$ denotes the ambient dimension, and all points and regions are considered in $\mathbb{R}^d$. We write $\mathbf{0}$ for the origin, e_1 for the first standard basis vector, and

$$B(x, r) := \{z \in \mathbb{R}^d : \|z - x\| \leq r\} \quad (2)$$

for the closed Euclidean ball of radius r centered at $x \in \mathbb{R}^d$, and

$$B^\circ(x, r) := \{z \in \mathbb{R}^d : \|z - x\| < r\} \quad (3)$$

for the corresponding open ball.

Definition 1 (Stepping-stone diversion neighborhood). Let $p, q \in \mathbb{R}^d$ be distinct, and set

$$\ell_{p,q} := \|q - p\|, \quad u_{p,q} := \frac{q - p}{\ell_{p,q}}. \quad (4)$$

For $\alpha \geq 1$, the stepping-stone diversion neighborhood determined by p and q is

$$S_{SS,\alpha}(p, q) := \{z \in \mathbb{R}^d : \|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha\}. \quad (5)$$

Its normalized region is

$$K_{SS,\alpha} := S_{SS,\alpha}(\mathbf{0}, e_1) = \{z \in \mathbb{R}^d : \|z\|^\alpha + \|z - e_1\|^\alpha \leq 1\}. \quad (6)$$

For later comparison, the closed and open Gabriel regions determined by p and q are the closed and open balls with diameter pq , respectively:

$$G_{cl}(p, q) := B\left(\frac{p+q}{2}, \frac{\ell_{p,q}}{2}\right), \quad G_{op}(p, q) := B^\circ\left(\frac{p+q}{2}, \frac{\ell_{p,q}}{2}\right). \quad (7)$$

The closed and open relative-neighborhood lunes are, respectively,

$$L_{cl}(p, q) := B(p, \ell_{p,q}) \cap B(q, \ell_{p,q}), \quad L_{op}(p, q) := B^\circ(p, \ell_{p,q}) \cap B^\circ(q, \ell_{p,q}). \quad (8)$$

For a finite or locally finite set $V \subset \mathbb{R}^d$ and distinct $p, q \in V$, these graphs are defined by

$$\begin{aligned} pq \in E(\text{GG}_{cl}(V)) &\iff G_{cl}(p, q) \cap (V \setminus \{p, q\}) = \emptyset, \\ pq \in E(\text{GG}_{op}(V)) &\iff G_{op}(p, q) \cap (V \setminus \{p, q\}) = \emptyset, \\ pq \in E(\text{RNG}_{cl}(V)) &\iff L_{cl}(p, q) \cap (V \setminus \{p, q\}) = \emptyset, \\ pq \in E(\text{RNG}_{op}(V)) &\iff L_{op}(p, q) \cap (V \setminus \{p, q\}) = \emptyset. \end{aligned} \quad (9)$$

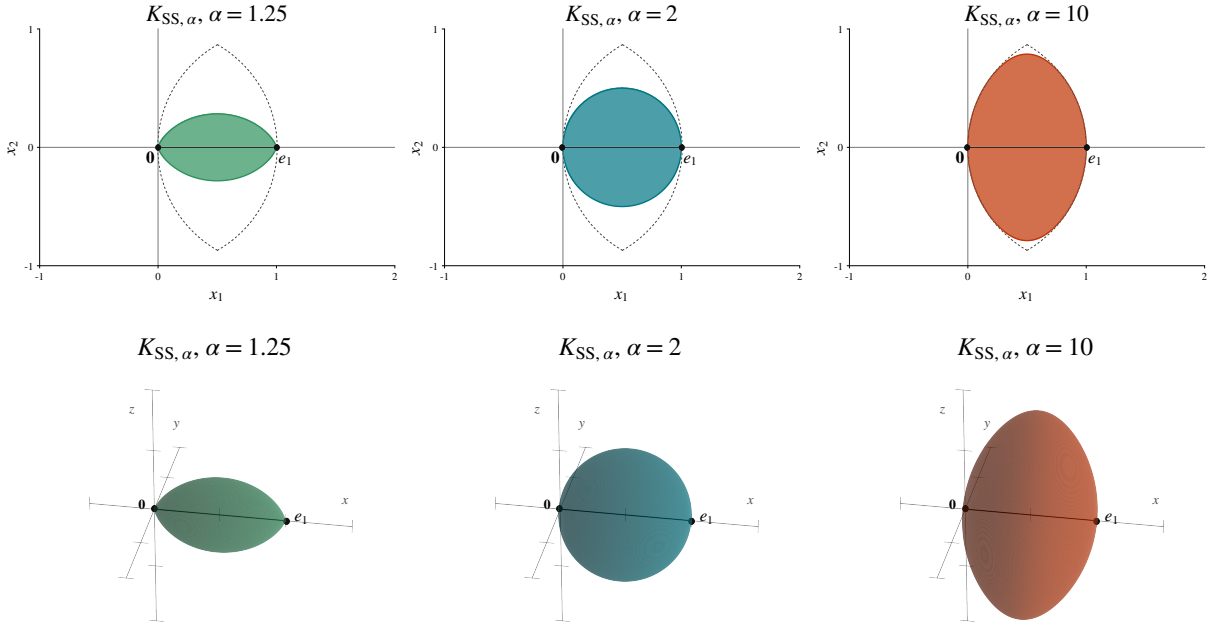


Figure 1: Normalized stepping-stone diversion neighborhoods for $\alpha = 1.25, 2$, and 10 . Top: planar regions, with the limiting relative-neighborhood lune shown by the dashed boundary. Bottom: three-dimensional regions for the same parameter values. The defining sites are fixed at $\mathbf{0}$ and e_1 .

Thus $\text{GG}_{\text{cl}}(V) \subseteq \text{GG}_{\text{op}}(V)$ and $\text{RNG}_{\text{cl}}(V) \subseteq \text{RNG}_{\text{op}}(V)$. Although the corresponding region boundaries have zero d -dimensional volume, the graph conventions may differ when a third site lies on such a boundary.

Throughout the paper, $\text{GG}(V)$ denotes $\text{GG}_{\text{cl}}(V)$ and $\text{RNG}(V)$ denotes $\text{RNG}_{\text{op}}(V)$, consistently with the original stepping-stone construction and the empty-region conventions of Kannangara et al. (2018); Cardinal et al. (2009). The alternative graphs $\text{GG}_{\text{op}}(V)$ and $\text{RNG}_{\text{cl}}(V)$ are always identified explicitly.

Definition 2 (Stepping-stone graph). For a finite or locally finite set $V \subset \mathbb{R}^d$, the stepping-stone graph with parameter α is the graph $\text{SSG}_{\alpha}(V) = (V, E_{\alpha}(V))$ such that two distinct points $p, q \in V$ are adjacent if

$$S_{\text{SS},\alpha}(p, q) \cap (V \setminus \{p, q\}) = \emptyset. \quad (10)$$

Proposition 1 (Similarity normalization). If Q is an orthogonal transformation satisfying $Qe_1 = u_{p,q}$, then

$$S_{\text{SS},\alpha}(p, q) = p + \ell_{p,q} Q K_{\text{SS},\alpha}. \quad (11)$$

The right-hand side is independent of the chosen transformation Q .

Proof. For $w \in \mathbb{R}^d$, substitute $z = p + \ell_{p,q} Qw$ in (5). The two normalized distances become $\|w\|$ and $\|w - e_1\|$, proving (11). If Q_1 and Q_2 both send e_1 to $u_{p,q}$, then $Q_2^{-1}Q_1$ fixes e_1 . The normalized region is invariant under every such orthogonal transformation, so $Q_1 K_{\text{SS},\alpha} = Q_2 K_{\text{SS},\alpha}$. $\square$

Thus it suffices to study the normalized region $K_{\text{SS},\alpha}$. Figure 1 shows the same three representative parameter values in two and three dimensions. The dashed planar boundary is the limiting relative-neighborhood lune.

To complement the static views in Figure 1, the computational companion (Espino-Montelongo and Maravillo, 2026) provides an interactive three-dimensional visualization of the normalized diversion neighborhood.

Proposition 2 (Elementary geometry). *For every $\alpha \geq 1$, the region $S_{SS,\alpha}(p, q)$ is compact and convex. It is centrally symmetric about*

$$m := \frac{p+q}{2} \quad (12)$$

and invariant under every Euclidean isometry fixing the line through p and q pointwise. If $\alpha > 1$, then $S_{SS,\alpha}(p, q)$ has nonempty interior and is strictly convex.

Proof. The function

$$F_{p,q,\alpha}(z) := \|z - p\|^\alpha + \|z - q\|^\alpha \quad (13)$$

is continuous and convex for $\alpha \geq 1$. Hence its sublevel set $S_{SS,\alpha}(p, q)$ is closed and convex. Its defining inequality implies

$$\|z - p\| \leq \ell_{p,q}, \quad \|z - q\| \leq \ell_{p,q}, \quad (14)$$

so the region is bounded and therefore compact.

The reflection $z \mapsto p + q - z$ exchanges the two distance terms and preserves the region, proving central symmetry. Every Euclidean isometry fixing the line through p and q pointwise preserves both distances and therefore preserves the region.

For $\alpha > 1$, the map $z \mapsto \|z - p\|^\alpha$ is strictly convex, and so is $F_{p,q,\alpha}$. If $z_1 \neq z_2$ belong to the region and $t \in (0, 1)$, then

$$F_{p,q,\alpha}(tz_1 + (1-t)z_2) < tF_{p,q,\alpha}(z_1) + (1-t)F_{p,q,\alpha}(z_2) \leq \ell_{p,q}^\alpha. \quad (15)$$

Thus the open segment joining z_1 and z_2 lies in the interior. Moreover,

$$F_{p,q,\alpha}(m) = 2 \left(\frac{\ell_{p,q}}{2} \right)^\alpha = 2^{1-\alpha} \ell_{p,q}^\alpha < \ell_{p,q}^\alpha, \quad (16)$$

so the interior is nonempty. $\square$

3. Endpoint and limiting regions

The stepping-stone family interpolates between three classical regions. We record the endpoint identities and the precise relative-neighborhood limit before studying the resulting graph filtration. All statements hold in every ambient dimension $d \geq 2$.

The graph identity at $\alpha = 2$ was established in the original planar stepping-stone study by Kannangara et al. (2018, Theorem 3.4). The following formulation keeps the boundary convention explicit and holds in every dimension.

Proposition 3 (Segment and Gabriel endpoints). *At $\alpha = 1$, the diversion neighborhood is the segment $[p, q]$. At $\alpha = 2$, it is the closed ball with diameter pq . Consequently, $SSG_2(V) = GG(V)$ for every finite or locally finite point set V . At $\alpha = 1$, two sites are adjacent precisely when the open segment joining them contains no other site; hence $SSG_1(V)$ is complete whenever no three sites are collinear.*

Proof. At $\alpha = 1$, the triangle inequality is an equality precisely on the segment joining p and q . At $\alpha = 2$, put $m = (p + q)/2$. The parallelogram identity gives

$$\|z - p\|^2 + \|z - q\|^2 = 2\|z - m\|^2 + 2 \left(\frac{\ell_{p,q}}{2} \right)^2, \quad (17)$$

so the defining inequality is equivalent to $\|z - m\| \leq \ell_{p,q}/2$. The graph statements follow from deleting the defining sites and from the closed Gabriel-ball convention. $\square$

The convergence of the planar graph to the relative-neighborhood graph was identified by Kannangara et al. (2018, Theorem 3.5). The next result states the corresponding region limit in arbitrary dimension.

Proposition 4 (Convergence to the relative-neighborhood lune). *The diversion neighborhoods increase from within to the open relative-neighborhood lune:*

$$\bigcup_{1 < \alpha < \infty} (S_{\text{SS},\alpha}(p, q) \setminus \{p, q\}) = L_{\text{op}}(p, q). \quad (18)$$

Proof. For $z \notin \{p, q\}$, the defining inequality implies

$$\|z - p\| < \ell_{p,q}, \quad \|z - q\| < \ell_{p,q}. \quad (19)$$

Thus every finite-parameter diversion neighborhood, after removal of its defining sites, is contained in $L_{\text{op}}(p, q)$.

Conversely, if $z \in L_{\text{op}}(p, q)$, then both normalized distances

$$\frac{\|z - p\|}{\ell_{p,q}}, \quad \frac{\|z - q\|}{\ell_{p,q}} \quad (20)$$

are strictly smaller than one. The sum of their α th powers tends to zero as $\alpha \rightarrow \infty$, so z belongs to the diversion neighborhood for every sufficiently large α . $\square$

4. Monotonicity and graph consequences

The region inclusion chains, connectivity, and spectrum formulation below hold in every ambient dimension. The intersection and Delaunay thresholds also hold in every dimension, while the candidate-count, tiling, dilation, and abstract-planarity statements identified as planar remain restricted to $\mathbb{R}^2$.

4.1. Strict region monotonicity

The edge monotonicity of the original planar family is due to Kannangara et al. (2018, Theorem 3.3). We first record the strict region refinement used throughout the paper.

Theorem 1 (Strict region monotonicity). *For every $1 \leq \alpha_1 < \alpha_2$ and every distinct $p, q \in \mathbb{R}^d$,*

$$S_{\text{SS},\alpha_1}(p, q) \subsetneq S_{\text{SS},\alpha_2}(p, q). \quad (21)$$

Proof. For $a, b \geq 0$, the map

$$r \mapsto (a^r + b^r)^{1/r}, \quad r > 0, \quad (22)$$

is nonincreasing. Applying this fact to $a = \|z - p\|$ and $b = \|z - q\|$ proves the non-strict inclusion.

For strictness, let $m = (p + q)/2$ and choose a unit vector v perpendicular to $q - p$. A point $m + tv$ belongs to $S_{\text{SS},\alpha}(p, q)$ precisely when

$$|t| \leq \ell_{p,q} \sqrt{2^{-2/\alpha} - \frac{1}{4}}. \quad (23)$$

The right-hand side is strictly increasing in α . Choosing t strictly between the radii corresponding to α_1 and α_2 gives a point in the larger region but not in the smaller. $\square$

4.2. Monotonicity and graph inclusions

The following order principle applies beyond the stepping-stone family. Let $K \subset \mathbb{R}^d$ contain the anchors $\mathbf{0}, e_1$ and be invariant under every orthogonal transformation fixing e_1 . For distinct p, q , define its pairwise unit-region image by

$$S_K(p, q) := p + \ell_{p,q} Q_{p,q} K, \quad Q_{p,q} e_1 = u_{p,q}. \quad (24)$$

The invariance condition makes this definition independent of the chosen orthogonal transformation. It is the natural arbitrary-dimensional analogue of the anchored-template construction of Cardinal et al. (2009) for this axially symmetric setting.

Let $G_K(V)$ join p and q precisely when $S_K(p, q)$ contains no site of $V \setminus \{p, q\}$. For any two such templates, region inclusion reverses edge inclusion:

$$K_1 \setminus \{\mathbf{0}, e_1\} \subseteq K_2 \setminus \{\mathbf{0}, e_1\} \implies G_{K_2}(V) \subseteq G_{K_1}(V). \quad (25)$$

Indeed, an empty copy of the larger region has an empty copy of the smaller region. The rest of the paper applies this elementary transfer principle to the nested templates $K_{\text{SS},\alpha}$.

Let $\text{MST}(V)$ denote a Euclidean minimum spanning tree of a **nonempty** finite point set V , and let $\text{Del}(V)$ denote its Delaunay graph (Mitchell and Mulzer, 2017). With the closed Gabriel-ball and open relative-neighborhood-lune conventions adopted above, the standard inclusions are

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{GG}(V) \subseteq \text{Del}(V). \quad (26)$$

The hierarchy is reviewed by Jaromczyk and Toussaint (1992); the closed-ball Gabriel convention and its Delaunay inclusion are explicit in Matula and Sokal (1980, Lemmas 1–2) and Cardinal et al. (2009). In particular, an empty closed diametral ball itself certifies the Delaunay empty-ball condition, including configurations with cospherical sites. In the planar setting, the graph nesting, the Gabriel and relative-neighborhood endpoints, and the Delaunay containment for parameters at least two were established by Kannangara et al. (2018, Theorems 3.3–3.5 and 3.11). The next proposition gives a region-based derivation that makes the open-closed conventions explicit and extends the resulting inclusion chains to every ambient dimension.

Proposition 5 (Monotonicity and classical proximity-graph inclusions). *If $1 \leq \alpha_1 < \alpha_2$, then*

$$E_{\alpha_2}(V) \subseteq E_{\alpha_1}(V). \quad (27)$$

Thus $\{\text{SSG}_\alpha(V) : \alpha \geq 1\}$ is a monotonically decreasing family of graphs under edge inclusion, although the inclusion need not be strict for a fixed point set. Furthermore, for every **nonempty** finite $V \subset \mathbb{R}^d$ and every finite $\alpha \geq 1$,

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{SSG}_\alpha(V). \quad (28)$$

If $1 \leq \alpha \leq 2$, then

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{GG}(V) \subseteq \text{SSG}_\alpha(V). \quad (29)$$

If $\alpha \geq 2$, then

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{SSG}_\alpha(V) \subseteq \text{GG}(V) \subseteq \text{Del}(V). \quad (30)$$

Proof. The first edge inclusion follows directly from Theorem 1 and the preceding implication; it is the region-inclusion formulation of the nestedness proved for the original d -spectrum by Kannangara et al. (2018, 2019).

For $z \notin \{p, q\}$, the inequality defining $S_{\text{SS},\alpha}(p, q)$ implies

$$\|z - p\| < \ell_{p,q}, \quad \|z - q\| < \ell_{p,q}. \quad (31)$$

Thus $S_{\text{SS},\alpha}(p, q) \setminus \{p, q\} \subset L_{\text{op}}(p, q)$, and the same implication gives $\text{RNG}(V) \subseteq \text{SSG}_\alpha(V)$. The remaining stepping-stone inclusions follow from Proposition 3 and Theorem 1; the other inclusions are classical. $\square$

Following the standard terminology of Jaromczyk and Toussaint (1992, Sec. II-C) and Eppstein (2002, Sec. 2), and the planar lune-based formulation of Maravillo Gómez, Calvillo Vives and Treviño Aguilar (2023, Def. 1), let $N_\beta(p, q)$ denote the closed region of influence of the β -skeleton. We use the closed-region convention throughout, so that the treatment of sites on the boundary is explicit. For $1 \leq \beta \leq 2$, the lune-based region is the intersection of the two closed balls of radius $\beta\ell_{p,q}/2$ centered at

$$c_1 = p + \frac{\beta}{2}(q - p), \quad c_2 = q + \frac{\beta}{2}(p - q). \quad (32)$$

For $0 < \beta \leq 1$, the circle-based region is the intersection of the two closed disks of radius $\ell_{p,q}/(2\beta)$ whose boundaries contain p and q . The two definitions agree at $\beta = 1$. At $\beta = 0$, we use the limiting region $N_0(p, q) = [p, q]$. The corresponding β -skeleton is denoted by $\beta\text{SG}_\beta(V)$ and is defined by

$$pq \in E(\beta\text{SG}_\beta(V)) \iff N_\beta(p, q) \cap (V \setminus \{p, q\}) = \emptyset. \quad (33)$$

Thus $\beta\text{SG}_1(V) = \text{GG}(V)$, while the lune-based 2-skeleton is the relative-neighborhood graph. The same endpoint terminology is used by Maravillo Gómez et al. (2023, Def. 1); our closed-region convention fixes the treatment of boundary sites as specified above.

Proposition 6 (Comparison with β -skeletons). *For $\alpha \geq 2$, set*

$$\beta_+(\alpha) := 2^{1-2/\alpha} \in [1, 2]. \quad (34)$$

Then, for every finite $V \subset \mathbb{R}^d$,

$$\text{RNG}(V) \subseteq \text{SSG}_\alpha(V) \subseteq \beta\text{SG}_{\beta_+(\alpha)}(V) \subseteq \text{GG}(V) \subseteq \text{Del}(V). \quad (35)$$

For $1 \leq \alpha \leq 2$, define

$$\rho_\alpha := \sqrt{2^{-2/\alpha} - \frac{1}{4}}, \quad \beta_-(\alpha) := \frac{4\rho_\alpha}{1 + 4\rho_\alpha^2} \in [0, 1]. \quad (36)$$

Then, for every finite planar point set $V \subset \mathbb{R}^2$,

$$\text{GG}(V) \subseteq \text{SSG}_\alpha(V) \subseteq \beta\text{SG}_{\beta_-(\alpha)}(V). \quad (37)$$

The endpoint values are

$$\beta_-(1) = 0, \quad \beta_-(2) = \beta_+(2) = 1. \quad (38)$$

For each fixed α , these values are best possible: in the respective parameter range, $\beta_-(\alpha)$ or $\beta_+(\alpha)$ is the largest β for which

$$N_\beta(p, q) \subseteq S_{\text{SS}, \alpha}(p, q) \quad (39)$$

holds for every pair p, q .

Proof. For $\alpha \geq 2$, the planar comparison with the lune-based β -skeleton is due to Kannangara et al. (2018, Theorem 3.10). The same calculation applies in $\mathbb{R}^d$. Normalize $p = \mathbf{0}$ and $q = e_1$, write $z = (x, Z)$ with $r = \|Z\|$, and put $a = \alpha/2 \geq 1$. By symmetry it is enough to take $0 \leq x \leq 1/2$. On the boundary of the lune-based region of influence $N_\beta(\mathbf{0}, e_1)$,

$$r^2 = \beta x - x^2. \quad (40)$$

Consequently, on that boundary the left-hand side of the normalized stepping-stone inequality is

$$f(x) := (\beta x)^a + (1 - (2 - \beta)x)^a. \quad (41)$$

For $\beta = \beta_+(\alpha)$, the convex function f has $f(0) = f(1/2) = 1$, and therefore $f(x) \leq 1$ on $[0, 1/2]$. Hence $N_{\beta_+(\alpha)}(\mathbf{0}, e_1)$ is contained in the stepping-stone region. Therefore, if the stepping-stone region contains no site of $V \setminus \{p, q\}$, neither does $N_{\beta_+(\alpha)}(p, q)$, which proves the middle inclusion in (35). The remaining inclusions follow from Proposition 5 and the monotonicity of β -skeletons: as β increases, the region of influence expands and the graph becomes sparser (Bose, Devroye, Evans and Kirkpatrick, 2006, Observation 1).

Now let $1 < \alpha < 2$ and consider a boundary point z of the normalized stepping-stone region. Put

$$u := \|z\|^\alpha, \quad 1 - u := \|z - e_1\|^\alpha, \quad b := \frac{2}{\alpha} \in (1, 2). \quad (42)$$

If $\theta = \angle \mathbf{0}ze_1$, the cosine law gives

$$\cos \theta = \frac{u^b + (1 - u)^b - 1}{2\{u(1 - u)\}^{b/2}}. \quad (43)$$

A direct differentiation shows that this symmetric expression is minimized at $u = 1/2$. Thus the largest boundary angle occurs in the midpoint slice and satisfies

$$\cos \theta_\alpha = 1 - 2^{2/\alpha-1}, \quad \sin \theta_\alpha = \beta_-(\alpha). \quad (44)$$

By the angle definition of the circle-based β -skeleton (Eppstein, 2002, Sec. 2), its region of influence is

$$\{z : \angle pqz \geq \pi - \arcsin(\beta_-(\alpha))\}. \quad (45)$$

To prove the containment, suppose that a point of this region were outside the stepping-stone region. Its longitudinal coordinate lies between those of p and q . Moving transversely toward $[p, q]$ at that fixed coordinate must first meet the stepping-stone boundary. The subtended angle strictly increases during this motion, producing a boundary angle larger than θ_α , contrary to the preceding maximum. Thus $N_{\beta_-(\alpha)}(p, q)$ is contained in the stepping-stone region. The edge inclusion follows again from the definition of an empty-region graph, and the first inclusion in (37) follows from Proposition 5. The endpoint cases $\alpha = 1, 2$ follow from the segment and closed Gabriel-ball cases.

It remains to prove that the parameters are best possible. In the hyperplane through the midpoint and perpendicular to pq , the maximum distance from the line pq is ρ_α for the stepping-stone region. The definitions of $\beta_-(\alpha)$ and $\beta_+(\alpha)$ make the corresponding region of influence reach exactly this distance. This distance is strictly increasing with β for both the circle-based and lune-based definitions. Hence every larger β gives a point of $N_\beta(p, q)$ outside the stepping-stone region, so (39) fails. $\square$

Remark 1 (Parameter values and consequences of inclusion). For example,

$$\beta_-(1.25) \approx 0.8568, \quad \beta_-(1.5) \approx 0.9656, \quad \beta_-(1.75) \approx 0.9946. \quad (46)$$

Every graph property closed under taking subgraphs passes from $\beta\text{SG}_{\beta_+(\alpha)}(V)$ to $\text{SSG}_\alpha(V)$, and deleting edges cannot decrease shortest-path distances. The inclusions alone do not imply connectivity or an upper bound on the spanning ratio, and the fact that a supergraph is dense or nonplanar gives no such conclusion for its subgraphs. Connectivity follows separately from $\text{MST}(V) \subseteq \text{SSG}_\alpha(V)$.

Corollary 1 (Planar embedding and Delaunay containment). *Let $V \subset \mathbb{R}^2$ be finite with $n := |V| \geq 3$, and let $\alpha \geq 2$. Then the geometric realization of $\text{SSG}_\alpha(V)$ is planar, and $\text{SSG}_\alpha(V)$ is a subgraph of the Delaunay graph. In particular, it has at most $3n - 6$ edges and is four-colorable.*

Proof. For $1 \leq \beta \leq 2$, the lune-based β -skeleton is planar and is a subgraph of the Delaunay graph (Jaromczyk and Toussaint, 1992; Maravillo Gómez et al., 2023). The conclusion follows from (35) because planarity and Delaunay containment are preserved under taking subgraphs. The edge bound and four-colorability are standard consequences of planarity. $\square$

Corollary 2 (Cyclic tilings). *Let $\mathcal{T} = \{T_1, T_2, \dots\}$ be a cyclic tiling of the plane, with discrete vertex set $V_{\mathcal{T}}$ and edge set $E_{\mathcal{T}}$. The empty-region definition extends without change from finite point sets to discrete point sets. For every finite $\alpha \geq 2$,*

$$E(\text{SSG}_{\alpha}(V_{\mathcal{T}})) \subseteq E_{\mathcal{T}}. \quad (47)$$

Thus every edge of the stepping-stone graph is an edge of the tiling.

Proof. For $\alpha \geq 2$, Proposition 6 gives

$$\text{SSG}_{\alpha}(V_{\mathcal{T}}) \subseteq \beta\text{SG}_{\beta_+(\alpha)}(V_{\mathcal{T}}), \quad 1 \leq \beta_+(\alpha) < 2. \quad (48)$$

By Maravillo Gómez et al. (2023, Theorem 1), if $\beta\text{SG}_{\beta_+(\alpha)}(V_{\mathcal{T}}) = (V_{\mathcal{T}}, E_{\beta})$, then $E_{\beta} \subseteq E_{\mathcal{T}}$. The result follows by transitivity. $\square$

Remark 2 (Delaunay starting graph in the plane). For $\alpha \geq 2$, Kannangara et al. (2018, Theorem 3.11 and Algorithm 1) use the Delaunay triangulation as the starting graph for computing the planar d -spectrum. In the notation used here, let $V \subset \mathbb{R}^2$ be a finite point set in Delaunay general position (no three sites collinear and no four sites cocircular), let $n = |V| \geq 3$, and let $\alpha \geq 2$. By (30), every edge of $\text{SSG}_\alpha(V)$ is an edge of $\text{Del}(V)$. Hence at most $3n - 6$ Delaunay edges need to be examined. The naive algorithm that compares every such edge with the remaining $n - 2$ sites takes $O(n^2)$ time; the linear number of Delaunay edges alone does not give a linear-time algorithm. For comparison, Lingas (1994) construct the planar RNG and lune-based β -skeletons from the Delaunay triangulation, while proximity algorithms in higher dimensions require different methods (Dickerson and Eppstein, 1996).

Remark 3 (Higher-dimensional Delaunay consequences). For $\alpha \geq 2$, Proposition 5 places $\text{SSG}_\alpha(V)$ inside the Delaunay graph in every dimension. Therefore, every graph property closed under taking subgraphs passes from $\text{Del}(V)$ to $\text{SSG}_\alpha(V)$. In particular, the shallow-minor bounds proved by Teng (1998) for Delaunay diagrams of random point sets also hold for the stepping-stone graph under the same random point model. This containment does not imply a linear number of edges in higher dimensions: Delaunay graphs can have $\Theta(n^2)$ edges in $\mathbb{R}^3$ (Agarwal and Matoušek, 1992), and the maximum number of edges in a Gabriel graph is also $\Theta(n^2)$ in dimension three and higher (Chazelle, Edelsbrunner, Guibas, Hershberger, Seidel and Sharir, 1994, Lemma 5.1).

Corollary 3 (Connectivity and planar geometric realization). *For every nonempty finite $V \subset \mathbb{R}^d$ and every finite $\alpha \geq 1$, the graph $\text{SSG}_\alpha(V)$ is connected. If $V \subset \mathbb{R}^2$ and $\alpha \geq 2$, its geometric realization is planar.*

Proof. The planar statements correspond to Kannangara et al. (2018, Lemmas 3.6–3.8). The connectivity assertion in arbitrary dimension follows because the graph contains a Euclidean minimum spanning tree for every finite $\alpha \geq 1$. If $V \subset \mathbb{R}^2$ and $\alpha \geq 2$, then $\text{SSG}_\alpha(V) \subseteq \text{GG}(V)$ by (30), and the planar Gabriel graph has a planar geometric realization. $\square$

For a connected geometric graph G on $V \subset \mathbb{R}^2$, write $d_G(p, q)$ for the Euclidean length of a shortest path from p to q in G . Its spanning ratio, also called its dilation, is

$$\text{dil}(G) := \max_{p \neq q \in V} \frac{d_G(p, q)}{\|p - q\|}. \quad (49)$$

Equivalently, G is a geometric t -spanner when $\text{dil}(G) \leq t$ (Cardinal et al., 2009; Bose and Smid, 2013).

Corollary 4 (Unbounded spanning ratio). *For every fixed $\alpha > 1$ and every $T > 0$, there exists a finite point set $V \subset \mathbb{R}^2$ such that $\text{SSG}_\alpha(V)$ is connected and*

$$\text{dil}(\text{SSG}_\alpha(V)) > T. \quad (50)$$

If $\alpha \geq 2$, its geometric realization is planar. Thus no fixed $\alpha > 1$ makes the stepping-stone graph a geometric t -spanner for all planar point sets, for any constant t . The restriction $\alpha > 1$ is essential: at $\alpha = 1$, every planar point set in general position yields the complete graph and hence dilation one.

Proof. For every $\alpha > 1$, Proposition 2 shows that the normalized stepping-stone template is convex and symmetric and has nonempty interior. Therefore Cardinal et al. (2009, Theorem 13) applies: for every anchored region with nonempty interior, they construct planar point sets whose spanning ratio is $\Omega(n^c)$ for some $c > 0$. Connectivity follows from Corollary 3; the same corollary gives a planar geometric realization when $\alpha \geq 2$. For this latter range, the conclusion can alternatively be obtained from the unbounded dilation of Gabriel graphs proved by Eppstein (2002), because $\text{SSG}_\alpha(V)$ is a connected subgraph of $\text{GG}(V)$. $\square$

Corollary 5 (Lower bounds on the spanning ratio). *The qualitative conclusion of Corollary 4 admits the following quantitative refinements.*

(i) *For every fixed finite $\alpha \geq 2$ and every $n \geq 2$, there exists a set V_n of n points in the plane such that*

$$\text{dil}(\text{SSG}_\alpha(V_n)) \geq \left(\frac{1}{2} - o(1)\right) \sqrt{n}. \quad (51)$$

(ii) *For every fixed $1 < \alpha < 2$, set*

$$\theta_\alpha := \frac{1}{3} \arcsin(\beta_-(\alpha)), \quad c_\alpha := \log_5 \left(\frac{5}{3 + 2 \cos \theta_\alpha} \right) > 0. \quad (52)$$

For every $n = 5^k + 1$, there exists a set V_n of n points in the plane for which

$$\text{dil}(\text{SSG}_\alpha(V_n)) = \Omega(n^{c_\alpha}). \quad (53)$$

The constants implicit in $o(1)$ and $\Omega(\cdot)$ may depend on the fixed parameter α .

Proof. If H is a connected spanning subgraph of a geometric graph G , then deleting the edges of $G \setminus H$ cannot shorten any graph distance; hence $\text{dil}(H) \geq \text{dil}(G)$. For $\alpha \geq 2$, apply this observation to

$$\text{SSG}_\alpha(V) \subseteq \beta\text{SG}_{\beta_+(\alpha)}(V) \quad (54)$$

and apply Bose et al. (2006, Theorem 3.2), which gives $(1/2 - o(1))\sqrt{n}$ for every $\beta \in [1, 2]$.

For $1 < \alpha < 2$, use instead $\text{SSG}_\alpha(V) \subseteq \beta\text{SG}_{\beta_-(\alpha)}(V)$. Here $\beta_-(\alpha) > 0$, and the construction of Eppstein (2002, Theorem 1), in the form recorded by Bose et al. (2006, Theorem 3.1), gives the displayed polynomial lower bound for every angle $0 < \theta < (2/3) \arcsin(\beta_-(\alpha))$. The stated choice of θ_α is admissible. In both cases the stepping-stone graph is connected by Corollary 3, so its dilation is finite and the subgraph comparison applies. $\square$

Corollary 6 (Uniform random points). *Fix a finite $\alpha \geq 2$, let $X_1, \dots, X_n$ be independent and uniform on $[0, 1]^2$, and put $V_n := \{X_1, \dots, X_n\}$. There are constants $c > 0$ and $0 < a < 1/12$ such that*

$$\mathbb{P} \left\{ \text{dil}(\text{SSG}_\alpha(V_n)) < c \sqrt{\frac{a \log n}{\log \log n}} \right\} \leq 2 \exp \left\{ -2n^{1-12a-o(1)} \right\}. \quad (55)$$

Consequently, $\text{dil}(\text{SSG}_\alpha(V_n)) \rightarrow \infty$ in probability.

Proof. For $\alpha \geq 2$, Proposition 5 gives $\text{SSG}_\alpha(V_n) \subseteq \text{GG}(V_n)$. The stepping-stone graph is connected almost surely, and therefore its dilation is at least that of the Gabriel graph. The probability bound is then exactly the random-point lower bound of Bose et al. (2006, Theorem 5.1). $\square$

A geometric graph on a point set has its edges drawn as straight-line segments, and two edges give rise to a crossing when they have an interior point in common (Ábrego, Fabila-Monroy, Fernández-Merchant, Flores-Peñaloza, Hurtado, Sacristán and Saumell, 2011). Crossings in higher-order nearest-neighbor, relative-neighborhood, Gabriel, and Delaunay graphs were studied by Ábrego et al. (2011); Bose, Collette, Hurtado, Korman, Langerman, Sacristán and Saumell (2013) studied the spanning ratio, diameter, connectivity, coloring, and crossing layers of higher-order Gabriel and Delaunay graphs. In the stepping-stone graph, the diversion neighborhood remains empty while its shape varies with α . The following results distinguish crossings in the geometric realization, Delaunay containment, and planarity of the graph.

Following Cardinal et al. (2009, Sec. 2), a planar anchored region $\mathcal{R}$ is obtained from a fixed template by translation, rotation, and uniform scaling. For $d \geq 2$, we use the natural higher-dimensional extension: the assigned regions are convex, symmetric in their two anchors, and equivariant under Euclidean similarities. We write $\text{ERG}_{\mathcal{R}}(V)$ for the empty region graph in which pq is an edge exactly when $\mathcal{R}(p, q)$ contains no point of $V \setminus \{p, q\}$.

Theorem 2 (The closed ball and intersection-free geometric realizations). *Let $d \geq 2$ and let $\mathcal{R}$ be a convex and symmetric anchored region in $\mathbb{R}^d$. For every finite $V \subset \mathbb{R}^d$, draw each edge of $\text{ERG}_{\mathcal{R}}(V)$ as the straight-line segment joining its endpoints. No two edges with disjoint endpoints intersect in their relative interiors, for every such V , if and only if*

$$G_{\text{cl}}(p, q) \subseteq \mathcal{R}(p, q) \quad (56)$$

for every pair of distinct points $p, q \in \mathbb{R}^d$. In the plane, this is equivalent to the planar-embedding characterization of Cardinal et al. (2009, Theorem 11). In particular, if $\mathcal{R}(p, q) \subsetneq G_{\text{cl}}(p, q)$ for some pair, then a four-point set contained in a two-dimensional affine subspace has an empty region graph with two edges whose relative interiors intersect.

Proof. Suppose first that $G_{\text{cl}}(p, q) \subseteq \mathcal{R}(p, q)$, then $\text{ERG}_{\mathcal{R}}(V)$ is a subgraph of the closed Gabriel graph. We show that two closed Gabriel edges with disjoint endpoints cannot intersect in their relative interiors in any dimension. If the two segments are collinear, an endpoint of one lies in the closed ball with diameter the other and excludes that edge. Otherwise, their endpoints lie in a common affine plane and form a convex quadrilateral whose diagonals are the two segments. At least one interior angle of the quadrilateral is at least $\pi/2$. By the diametral-ball criterion, the vertex of that angle lies in the closed Gabriel ball of the opposite diagonal, excluding one of the two edges.

Conversely, suppose that (56) fails for some p, q . Choose $a \in G_{\text{cl}}(p, q) \setminus \mathcal{R}(p, q)$. Convexity and the anchoring of p and q imply that a is not on the line through the two anchors. Let P be the two-dimensional affine plane containing p, q , and a . The sections

$$G_{\text{cl}}(x, y) \cap P, \quad \mathcal{R}(x, y) \cap P, \quad x, y \in P, \quad (57)$$

form, respectively, the closed diametral disk and a convex symmetric anchored region in P . The latter does not contain the former for the pair p, q . The planar tight-region construction of Cardinal et al. (2009, Theorem 11) therefore gives four points in P whose empty-region graph contains two edges crossing in their relative interiors. The same four points, viewed as a subset of $\mathbb{R}^d$, give the required counterexample. This proves necessity and the final assertion. $\square$

Corollary 7 (Open Gabriel and stepping-stone regions). *For every $d \geq 2$, every pair of distinct points $p, q \in \mathbb{R}^d$, and every $1 \leq \alpha < 2$,*

$$S_{\text{SS}, \alpha}(p, q) \setminus \{p, q\} \subseteq G_{\text{op}}(p, q). \quad (58)$$

Consequently, for every finite $V \subset \mathbb{R}^d$,

$$\text{GG}_{\text{op}}(V) \subseteq \text{SSG}_{\alpha}(V), \quad 1 \leq \alpha < 2. \quad (59)$$

The open Gabriel graph, and therefore the stepping-stone graph in this parameter range, can have crossing edges. In contrast, for every $\alpha \geq 2$, the geometric realization of $\text{SSG}_{\alpha}(V)$ has no crossing between edges with disjoint endpoints. Hence the value $\alpha = 2$ is best possible in every dimension.

Proof. Let $z \in S_{SS,\alpha}(p, q) \setminus \{p, q\}$ and set

$$r := \frac{\|z - p\|}{\ell_{p,q}}, \quad s := \frac{\|z - q\|}{\ell_{p,q}}. \quad (60)$$

Then $r, s \in (0, 1)$ and $r^\alpha + s^\alpha \leq 1$. Since $1 \leq \alpha < 2$,

$$r^2 + s^2 < r^\alpha + s^\alpha \leq 1. \quad (61)$$

The strict inequality $r^2 + s^2 < 1$ is precisely the condition that z belong to the open ball with diameter pq , which proves (58). Because an edge is present when its region contains no other site, this containment gives (59).

The failure of a planar embedding is already visible on the point set

$$V := \{\mathbf{0}, e_1, e_2, e_1 + e_2\} \subset \{0, 1\}^d. \quad (62)$$

The segments joining $\mathbf{0}$ to $e_1 + e_2$ and e_1 to e_2 are the two diagonals of a unit square and cross at their midpoint. For either diagonal, the other two sites lie on the boundary of its Gabriel ball. Thus both diagonals are edges of $\text{GG}_{\text{op}}(V)$ and, by (59), of $\text{SSG}_\alpha(V)$.

Finally, if $\alpha \geq 2$, then $\text{SSG}_\alpha(V) \subseteq \text{GG}_{\text{cl}}(V)$ by (30). Applying Theorem 2 to the closed ball shows directly, in every dimension, that no two edges with disjoint endpoints intersect in their relative interiors. $\square$

Theorem 3 (Delaunay containment). *Let $d \geq 2$. The inclusion*

$$\text{SSG}_\alpha(V) \subseteq \text{Del}(V) \quad (63)$$

holds for every finite $V \subset \mathbb{R}^d$ if and only if $\alpha \geq 2$. For every $1 \leq \alpha < 2$, there is a full-dimensional point set for which the inclusion fails; after translation and rescaling, the point set may be chosen with coordinates in $\mathbb{N}_0$. Thus the value $\alpha = 2$ is best possible.

Proof. The planar inclusion for $\alpha \geq 2$ was proved by Kannangara et al. (2018, Theorem 3.11). In arbitrary dimension it follows from the region inclusion in (30). It remains to show that the parameter range is best possible and that failure occurs in every dimension. Fix $1 \leq \alpha < 2$ and define

$$\rho_\alpha := \sqrt{2^{2-2/\alpha} - 1}. \quad (64)$$

Then $0 \leq \rho_\alpha < 1$. Choose t with $\rho_\alpha < t < 1$ and let

$$V_{d,t} := \{-e_1, e_1\} \cup \{\pm te_j : 2 \leq j \leq d\}. \quad (65)$$

For $p = -e_1$, $q = e_1$, and every other site $z \in V_{d,t}$,

$$\|z - p\| = \|z - q\| = \sqrt{1 + t^2}, \quad (66)$$

and the choice $t > \rho_\alpha$ gives

$$2(1 + t^2)^{\alpha/2} > 2^\alpha = \|p - q\|^\alpha. \quad (67)$$

Thus none of the other sites belongs to $S_{SS,\alpha}(p, q)$, and pq is a stepping-stone edge.

Every sphere through p and q has a center of the form $c = \sum_{j=2}^d u_j e_j$ and squared radius $1 + \|c\|^2$. Choose an index $j \geq 2$ and a sign $s \in \{-1, 1\}$ such that $su_j \geq 0$. Then

$$\begin{aligned} \|c - ste_j\|^2 &= \|c\|^2 + t^2 - 2stu_j \\ &\leq \|c\|^2 + t^2 < \|c\|^2 + 1. \end{aligned} \quad (68)$$

Hence every sphere through p and q contains another site of $V_{d,t}$ in its interior. Therefore no empty sphere passes through p and q , so $pq \notin E(\text{Del}(V_{d,t}))$.

For an integer-coordinate version, choose positive integers M and k with $\rho_\alpha M < k < M$, replace e_1 by Me_1 and te_j by ke_j , and translate the resulting finite set into $\mathbb{N}_0^d$. The same strict inequalities prove the claim and also show that the counterexample persists under all sufficiently small perturbations. $\square$

Corollary 8 (Planarity of the graph). *For point sets in the plane, every $\text{SSG}_\alpha(V)$ with $\alpha \geq 2$ is planar, whereas for every $1 \leq \alpha < 2$ there is a finite planar point set whose stepping-stone graph is nonplanar. Thus the value $\alpha = 2$ is best possible for planarity.*

In contrast, for every $d \geq 3$ there is a fixed finite set $V_d \subset \mathbb{N}_0^d$ such that $\text{SSG}_\alpha(V_d)$ is nonplanar for every finite $\alpha \geq 1$. Consequently, no value of α guarantees planarity in dimensions at least three.

Proof. When $d = 2$ and $\alpha \geq 2$, planarity follows from the planar geometric realization in Corollary 7. When $1 \leq \alpha < 2$, the closed ball is a tight region for planarity by Cardinal et al. (2009, Theorem 24). Since $S_{\text{SS},\alpha}(p, q) \subsetneq G_{\text{cl}}(p, q)$, their tightness construction supplies a finite planar point set whose stepping-stone graph is nonplanar.

For $d \geq 3$, take

$$V_d := \{0, 1, 2\}^3 \times \{0\}^{d-3}. \quad (69)$$

If $p, q \in V_d$ are lattice neighbors, then $\|p - q\| = 1$, while every third site $z \in V_d \setminus \{p, q\}$ satisfies $\|z - p\| \geq 1$ and $\|z - q\| \geq 1$. Therefore,

$$\|z - p\|^\alpha + \|z - q\|^\alpha \geq 2 > 1 = \|p - q\|^\alpha, \quad (70)$$

so every nearest-neighbor lattice pair is a stepping-stone edge for every $\alpha \geq 1$. Thus $\text{SSG}_\alpha(V_d)$ contains the graph of the $3 \times 3 \times 3$ grid. This bipartite graph has 27 vertices and

$$3(3 - 1)3^2 = 54 \quad (71)$$

edges. A simple planar bipartite graph on 27 vertices has at most $2(27) - 4 = 50$ edges, so the grid, and hence $\text{SSG}_\alpha(V_d)$, is nonplanar. $\square$

Remark 4 (Binary point set in dimensions $d \geq 4$). The use of the coordinate value 2 in (69) is needed only to cover dimension three with this elementary counting argument. For every $d \geq 4$, the pure binary set $V = \{0, 1\}^d$ already works. All unit Hamming-neighbor pairs are stepping-stone edges for every $\alpha \geq 1$, so the graph contains the four-dimensional hypercube graph Q_4 . This bipartite graph has 16 vertices and $32 > 2(16) - 4 = 28$ edges and is therefore nonplanar.

Remark 5 (Comparison with higher-order proximity graphs). The crossings above arise because the stepping-stone region is smaller than the Gabriel region when $\alpha < 2$. This differs from the higher-order construction studied by Ábrego et al. (2011) and Bose et al. (2013): in an order- k proximity graph, the region is fixed and may contain at most k points of V different from its defining sites. In the stepping-stone graph the region remains empty and its shape varies with α . For this family, $\alpha = 2$ is best possible, in every dimension, for a crossing-free geometric realization.

Remark 6 (Tight regions in the plane). In the framework of anchored regions studied by Cardinal et al. (2009), the closed ball is the unique tight region that ensures a planar embedding, it is also a tight region for planarity, and the open lune is the unique tight region for connectivity. For $\alpha \geq 2$, the stepping-stone region lies between these two regions:

$$G_{\text{cl}}(p, q) \subseteq S_{\text{SS},\alpha}(p, q), \quad S_{\text{SS},\alpha}(p, q) \setminus \{p, q\} \subseteq L_{\text{op}}(p, q). \quad (72)$$

The inclusions in Proposition 5 therefore give connectivity and a planar geometric realization in the plane. The tight-region theorems of Cardinal et al. (2009) are planar; the result in Corollary 7 extends to every dimension because two crossing segments lie in a common affine plane. In dimensions at least three, a crossing-free geometric realization does not imply planarity of the graph, as the lattice construction in Corollary 8 demonstrates.

4.3. α -values and the stepping-stone spectrum

Kannangara et al. define the d -value of an edge as the smallest real configuration value for which its diversion neighborhood contains another site, and call the set of edges labeled with their d -values the d -spectrum (Kannangara et al., 2018, Def. 3.12). Their Algorithm 1 uses the Delaunay triangulation as a starting graph and assigns d -value ∞ when the open lune is empty. We use the same terminology, replacing the configuration value d by α . The equation below defines each α -value, extends the construction to $\mathbb{R}^d$, bounds the number of distinct graphs, and identifies the value after which the graph is the relative-neighborhood graph defined by the open lune.

For distinct $p, q, z \in V$, define

$$r_{p,q}(z) := \frac{\|z - p\|}{\ell_{p,q}}, \quad s_{p,q}(z) := \frac{\|z - q\|}{\ell_{p,q}}. \quad (73)$$

If $r_{p,q}(z) < 1$ and $s_{p,q}(z) < 1$, let $\tau_{p,q}(z)$ be the unique number in $[1, \infty)$ satisfying

$$r_{p,q}(z)^{\tau_{p,q}(z)} + s_{p,q}(z)^{\tau_{p,q}(z)} = 1. \quad (74)$$

If either normalized distance is at least one, set $\tau_{p,q}(z) := \infty$. Finally, define the α -value of the pair by

$$\tau_V(p, q) := \min_{z \in V \setminus \{p, q\}} \tau_{p,q}(z), \quad (75)$$

where the minimum of the empty set is interpreted as ∞ .

Theorem 4 (The α -spectrum and the relative-neighborhood graph). *Let $V \subset \mathbb{R}^d$ be finite. The finite values in (74) are well defined, and for every finite $\alpha \geq 1$,*

$$pq \in E_\alpha(V) \iff \alpha < \tau_V(p, q). \quad (76)$$

Consequently, every edge is removed at most once, the graph is constant between consecutive finite α -values, and the family contains at most $\binom{|V|}{2} + 1$ distinct graphs. Moreover,

$$\tau_V(p, q) = \infty \iff pq \in E(\text{RNG}_{\text{op}}(V)). \quad (77)$$

If

$$\alpha_*(V) := \max \{ \tau_V(p, q) : \tau_V(p, q) < \infty \}, \quad (78)$$

with $\alpha_*(V) = 1$ when the set is empty, then

$$\text{SSG}_\alpha(V) = \text{RNG}_{\text{op}}(V) \quad \text{for every } \alpha \geq \alpha_*(V). \quad (79)$$

Proof. Suppose first that $r := r_{p,q}(z) < 1$ and $s := s_{p,q}(z) < 1$. The triangle inequality gives $r + s \geq 1$. The function

$$\alpha \mapsto r^\alpha + s^\alpha \quad (80)$$

is continuous and strictly decreasing on $[1, \infty)$, has value at least one at $\alpha = 1$, and tends to zero. Hence the equation $r^\alpha + s^\alpha = 1$ has a unique solution $\tau_{p,q}(z) \in [1, \infty)$. The defining inequality of the diversion region gives

$$z \in S_{\text{SS}, \alpha}(p, q) \iff \alpha \geq \tau_{p,q}(z). \quad (81)$$

If either $r \geq 1$ or $s \geq 1$, then z cannot belong to any finite-parameter diversion neighborhood, consistently with $\tau_{p,q}(z) = \infty$.

The pair pq is an edge precisely when no site in $V \setminus \{p, q\}$ belongs to its diversion neighborhood. Taking the minimum over these sites proves (76). Since V has only finitely many pairs, only finitely many finite α -values occur, and each edge can be removed only once.

Finally, $\tau_V(p, q) = \infty$ precisely when no third site has both normalized distances strictly below one. This is exactly the condition that $L_{\text{op}}(p, q)$ contain no point of $V \setminus \{p, q\}$, so these pairs are exactly the edges of $\text{RNG}_{\text{op}}(V)$. Once all finite α -values have been reached, only these edges remain, proving (79). $\square$

Remark 7 (Dependence on the point set). The finite value $\alpha_*(V)$ necessarily depends on V and cannot be bounded uniformly over all finite point sets. Take $p = (0, 0)$, $q = (1, 0)$, and $z = (1/2, h)$, where $0 < h < \sqrt{3}/2$, and write $r_h = \sqrt{1/4 + h^2} < 1$. Then $z \in L_{\text{op}}(p, q)$ and it first enters $S_{\text{SS}, \alpha}(p, q)$ when

$$2r_h^\alpha \leq 1, \quad \text{equivalently} \quad \alpha \geq \frac{\log 2}{-\log r_h}. \quad (82)$$

As $h \uparrow \sqrt{3}/2$, this value tends to infinity.

5. Implementation

An earlier implementation of the stepping-stone graph was developed in Java 8 by Kannangara et al. (2018, Sec. 4.2). For planar input, that implementation constructs the Delaunay triangulation, computes the d -spectrum, approximates edge d -values numerically with the secant method, and extracts the requested stepping-stone graph. It was used in their visualization and movement-analysis system.

The stepping-stone graph is also available in Python through PROXIMITYGRAPHS, an open-source package for constructing and analyzing proximity graphs (Maravillo, Villarreal de la Cerda and Espino-Montelongo, 2026). The package provides reusable routines for constructing stepping-stone diversion neighborhoods and graphs within Python scientific-computing workflows. It is distributed under the MIT License and is available from PyPI; it can be installed with `pip install proximitygraphs`. The source is maintained in the public GitHub repository, and version 0.1.0a2 is archived on Zenodo under DOI 10.5281/zenodo.21088183. This Python implementation complements the original Java system and makes the graph family available together with other classical proximity graphs.

6. Conclusion

The stepping-stone neighborhoods form a strictly increasing family from the segment, through the closed Gabriel ball at $\alpha = 2$, toward the open relative-neighborhood lune. Because empty-region inclusion reverses edge inclusion, this produces a decreasing graph filtration in every ambient dimension. The resulting chains imply connectivity and place the family relative to the Gabriel, relative-neighborhood, Delaunay, and β -skeleton graphs.

The value $\alpha = 2$ separates two universal geometric regimes. For $\alpha \geq 2$, every straight-line realization has no proper intersection between edges with disjoint endpoints and every edge belongs to the Delaunay graph. For $1 \leq \alpha < 2$, fixed planar witnesses show that neither property is guaranteed. These thresholds remain valid in every dimension because a proper intersection of two segments is contained in an affine plane, while the Delaunay counterexample admits a full-dimensional construction.

Abstract planarity requires a different conclusion. In the plane, $\alpha = 2$ is the sharp universal threshold, consistently with the tight-region theory of Cardinal et al. (2009). In dimensions at least three, intersection-free realization does not imply abstract planarity: integer-coordinate point sets yield nonplanar stepping-stone graphs for every parameter. This separates a geometric property of the given realization from a combinatorial property of the graph.

Finally, the site-entry equation extends the stepping-stone spectrum of Kannangara et al. (2018) to arbitrary dimension. For every finite point set, each edge is removed at most once, at most $\binom{|V|}{2} + 1$ distinct graphs occur, and the filtration stabilizes at the open relative-neighborhood graph after the largest finite threshold. These conclusions provide a finite combinatorial description of a continuously parameterized proximity-graph family.

CRedit authorship contribution statement

Heriberto Espino-Montelongo: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Visualization, Writing – original draft. **Héctor Maravillo:** Investigation, Supervision, Visualization, Writing – review and editing.

Code and data availability

No external datasets were used. A reusable Python implementation of the stepping-stone graph is available in the PROXIMITYGRAPHS package on PyPI and in its public source repository. Version 0.1.0a2 is archived on Zenodo under DOI 10.5281/zenodo.21088183 (Maravillo et al., 2026).

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation, the authors used ChatGPT by OpenAI for language, wording, concept identification, and code formatting. The authors reviewed and verified all resulting material and take full responsibility for the manuscript. All figures were generated deterministically from mathematical formulas and numerical data using author-verified code; no generative image model was used.

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